

# MATH 122 Fall 2026-2027

It is my prerogative as your instructor to change the course during the term at my discretion. Any changes will be communicated via Canvas and/or Drexel email, so be sure to check it frequently. Course announcements, assessment information, and other details will be regularly posted to Canvas. You are expected to be fully aware of the following policies and expectations, so review this information carefully and ask me if you have further questions.

## Prerequisites

Students should have obtained ONE of the following:

- MATH 117 Minimum Grade: C-
- MATH 121 Minimum Grade: C-
- AP Calculus AB Minimum Grade: 4
- AP Calculus AB Subscore Grade Minimum Grade: 4

## Course Instructor

Peter Mao (Office: Korman Center 221, Email: [pm929@drexel.edu](mailto:pm929@drexel.edu))

## Class Meetings

Monday through Thursday (MTWR), 4:00–4:50 PM, in One Drexel Plaza GL33.

## Office Hours

Office hours will be held in the Math Resource Center (MRC), located in Korman 207, at the following times:

- Monday 6:00–7:00 PM
- Thursday 12:00–2:00 PM

I encourage you to use the MRC's services in general, not just when I am holding my office hours. See the [Tutoring Services](#) section of this syllabus for more details.

## Course Description and Expectations

The subject matter of the course is integration of single-variable functions, including various integration techniques and applications. Upon successful completion of Math 122, you should be able to

- Understand the Fundamental Theorem of Calculus and how it relates differentiation and integration.
- Demonstrate a proficiency in various integration techniques – such as: substitution, parts, and partial fractions.
- Understand the definition and interpretations of the definite integral.
- Use definite integrals to solve physical problems, such as: area, volume, arc length of a planar curve, and work.
- Be able to solve separable ordinary differential equations using the technique of separation of variables.
- Demonstrate knowledge of a polar coordinate system, including the techniques for and applications of differentiating and integrating polar functions.

## Calculator

A non-graphing and non-programmable scientific calculator (for example, [TI-30XIISTM](#)) will be allowed on all quizzes and exams. Use of a graphing/programmable calculator or cell phone during a quiz or exam will be considered a violation of the University academic integrity policy.

## Textbook Suggestions

There is no required text for the course. The recommended text is *Calculus: Early Transcendentals*, by Anton, Bivens, and Davis. Other acceptable textbooks include those written by (in no particular order): Stewart; Briggs, Cochran, and Gillett; Rogawski; Larson and Edwards. Basically, your book's table of contents should include all of the topics listed in the course schedule on the last page of this syllabus.

In addition, you may consider the following supplementary resources:

- Paul's Online Notes, with relevant topics in his "Calculus I" and "Calculus II" courses:  
<https://tutorial.math.lamar.edu/Classes/CalcI/CalcI.aspx>  
<https://tutorial.math.lamar.edu/Classes/CalcII/CalcII.aspx>
- Dr. Dimitrios Papadopoulos's notes:  
<https://www.math.drexel.edu/~dp399/math122/index.html>
- Dr. Trefor Bazett's video lectures:  
<https://www.youtube.com/playlist?list=PLHXZ90QGMqxfT9RMcReZ4WcoVILP4k6-m>  
<https://www.youtube.com/playlist?list=PLHXZ90QGMqxc4ySKTIW19TLrT91Ik9M4n>
- Khan Academy:  
<https://www.khanacademy.org/math/integral-calculus/ic-integration>

## Attendance

Attendance is essential for success in this course. You are responsible for all the material discussed in class, even if you are not present. The quarter system moves very quickly – if you don't do your work regularly, it is easy to fall behind.

## Practice Problems and Extra Credit Assignments

<http://www.math.drexel.edu/classes/Calculus/math122.html>

For the topics covered in class, practice problem sets have been crafted to illustrate important concepts and techniques that you are expected to master. These problems can be found via the above link. They should be worked regularly and in detail. It is only by doing the problems yourself that you will acquire the skills needed for proficiency in the course. It is your responsibility to look at all of the problems, do the work, and check the answers.

These practice problems will be grouped into weekly extra credit assignments. You can submit your work on these problems via Canvas on a weekly basis by the specified deadlines. They will count up to 5% extra credit towards your course grade.

For each extra credit assignment, your submission MUST be a single PDF file of your handwritten work. Submissions in any other format will NOT receive any extra credit points. If you do not have a scanner, there are free phone apps that you can use, for example:

<https://pdf.wondershare.com/mobile-app/free-pdf-scanner-app.html>

Your file MUST be readable with proper orientation for all pages. Each extra credit assignment will ONLY be checked for completeness and will be graded based on the following standard:

- 5 pts: [90%, 100%] complete
- 4 pts: [70%, 90%) complete
- 3 pts: [50%, 70%) complete
- 2 pts: [30%, 50%) complete
- 1 pts: [20%, 30%) complete
- 0 pts: [0%, 20%) complete

For a problem to be counted as completed, all work for the problem must be shown (EXCEPT for graphical problems). Any problem without relevant work shown will NOT be counted as completed. If a problem has multiple parts, each part will be checked as an individual problem. Since these assignments are for extra credit, NO late submission will be accepted. No exceptions.

## In-Class Quizzes

There will be six in-class quizzes. Dates and coverage of quizzes will be announced on Canvas. A quiz missed due to participation in a university-sponsored athletic competition or conference may be rescheduled or made up; please let me know in advance. Otherwise, missing a quiz for any reason will earn a 0 for that quiz. I recognize that a student might miss a quiz due to a health

reason or other unforeseen emergency. To accommodate such eventualities, every student will have their two lowest quiz grades dropped.

## Exams

There will be two midterm exams during the term on the Mondays of Weeks 5 and 9:

**Exam 1:** Monday, October 19

**Exam 2:** Monday, November 16

An announcement about the exact coverage and location of each midterm exam will be posted on Canvas at least one week prior to the exam date.

A single makeup exam covering the material on both midterms will be offered during the last week of the course to students who miss one of the midterm exams. Anyone missing a midterm exam who fails to take the makeup exam will be given a score of 0 for the exam. Note that the makeup may not count for both midterms, so if both midterm exams are missed, then you will earn a 0 for one of the exams. Exams may not be retaken for any reason.

There will be a comprehensive two-hour final exam scheduled during the final exam week (Monday, 12/07/2026 – Saturday, 12/12/2026) at the end of the term. The exact date and time are to be determined by the registrar. **Do not make travel plans until after the announcement of the final exam is made. You are expected to take the final exam at the time scheduled by the registrar!**

**All quizzes and exams are closed book and closed notes. Except for a non-graphing and non-programmable scientific calculator, any other types of calculators or electronic devices are not permitted for quizzes and exams. Using such a device during a quiz or exam will be considered a violation of the university's academic honesty policy.**

## Course Grading

|                                         | Option 1 | Option 2 |
|-----------------------------------------|----------|----------|
| <b>Quizzes</b>                          | 22%      | 22%      |
| <b>Lower Midterm</b>                    | 24%      | 14%      |
| <b>Higher Midterm</b>                   | 24%      | 24%      |
| <b>Final Exam</b>                       | 30%      | 40%      |
| <b>Practice Problems (Extra Credit)</b> | 5%       | 5%       |

Course totals will be computed in the two different ways shown above. The higher score “ $x$ ” of these two options will be converted to a letter grade based on the following cutoffs. Course totals will not be rounded before these cutoffs are applied.

|                      |                      |                      |
|----------------------|----------------------|----------------------|
| A-                   | A                    | A+                   |
| $90\% \leq x < 93\%$ | $93\% \leq x < 97\%$ | $97\% \leq x$        |
| B-                   | B                    | B+                   |
| $80\% \leq x < 83\%$ | $83\% \leq x < 87\%$ | $87\% \leq x < 90\%$ |
| C-                   | C                    | C+                   |
| $70\% \leq x < 73\%$ | $73\% \leq x < 77\%$ | $77\% \leq x < 80\%$ |
| F                    | D                    | D+                   |
| $0\% \leq x < 60\%$  | $60\% \leq x < 65\%$ | $65\% \leq x < 70\%$ |

## Tutoring Services

Students can receive extra assistance from the Math Resource Center (MRC). Hours and more details can be found at: <https://drexel.edu/coas/academics/departments-centers/mathematics/math-resource-center/>

The Center for Learning and Academic Success Services offers both peer and professional academic coaching. Academic coaches work with students on any topic related to being successful academically, including study skills, time management, navigating Drexel, organization skills, test-taking, etc. Students will be encouraged to set goals and work towards accomplishing them through participation in a combination of individual coaching, identification and utilization of resources, and targeted action planning. More information is available at <https://drexel.edu/studentlife/student-success/academic-resource-center/find-resources/center-learning-academic-success/coaching>

## Disabilities and Accommodations

Students requesting accommodations due to a disability at Drexel University need to request a current Accommodations Verification Letter (AVL) in the Clockwork database before accommodations can be made. These requests are received by Disability Resources (DR), who then issues the AVL to the appropriate contacts. For additional information, visit the DR website.

- DR Website: <https://drexel.edu/disability-resources/>
- Clockwork Database: <https://drexel.edu/disability-resources/support-accommodations/student-family-resources/>
- DR Phone Number: 215.895.1401
- DR E-mail Address: [disability@drexel.edu](mailto:disability@drexel.edu)

## Academic Honesty

Cheating and other forms of academic misconduct are serious offenses and are dealt with harshly, e.g. at the very least a 0 on an exam and a letter sent to the Office of Student Conduct. Students should be familiar with the following policies:

<https://drexel.edu/studentlife/student-conduct/code-of-conduct/academic-integrity-policy>  
<https://drexel.edu/studentlife/community-standards/code-of-conduct>

## Use of AI Tools

You may use AI tools (e.g., ChatGPT, Claude) and online solvers (e.g., Symbolab, Wolfram Alpha) to help you learn: asking for an explanation of a concept, or checking a final answer after you have worked a problem yourself. Work you submit, including extra credit assignments, must reflect your own reasoning. Quizzes and exams are timed, and no notes or tools other than a scientific calculator are allowed, so success on them requires both a fluent understanding of the material and the speed that comes from practice. The only way to build either is by working problems on your own. Keep in mind that the reliability of AI tools varies widely depending on how you use them. Using a chat interface alone is the least reliable, as its answers occasionally contain small errors. Agentic modes, in which you prompt the tool to check its own work by writing code, formalizing the mathematics, or running additional instances of itself, are much more reliable. However, any errors that do slip through will be that much harder to detect unless you have a strong understanding of the mathematics yourself.

## Course Drop & Withdrawal Policies

Students should be familiar with the following policies:

<https://drexel.edu/drexelcentral/registration/courses/adding-dropping>  
<https://drexel.edu/drexelcentral/registration/courses/course-withdraw>

Students should also keep in mind the following important dates:

|                    |                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------|
| Monday 09/28, 2026 | Last day to Add/Drop a course, either with Academic Advisor assistance or via DrexelOne by 11:59 PM |
| Monday 11/09, 2026 | Last day to Withdraw from a course with Academic Advisor assistance                                 |

## Incomplete (INC) Grade Policy

To be eligible to potentially receive a grade of incomplete for the course, a student must be unable to finish the course by the end of the quarter due to extreme circumstances that are beyond their control (documentation may be required) and BOTH of the following must be true:

1. At least 70% of course work must be completed with a passing grade of C or better on the completed work.
2. A formal incomplete grade contract must be filled out and signed by both the instructor and the student before the grades are submitted for the course.

In cases where incompletes are given, only missing work that does not violate the makeup policies on the syllabus can be made up and no work may be repeated (exams and quizzes may not be

retaken, and assignments may not be redone).

More details can be found at <https://drexel.edu/registrar/grades>

## Grade Appeals Policy

<https://drexel.edu/provost/policies-calendars/student-academic-concerns-complaints>

<https://drexel.edu/drexelcentral/records/grades/academic-standing>

## Appropriate Use of Course Materials

It is important to recognize that some or all of the course materials provided to you are the intellectual property of Drexel University, the course instructor, or others. Use of this intellectual property is governed by Drexel University policies, including the IT-1 policy found at: <https://drexel.edu/it/about/policies/policies/01-Acceptable-Use/>

Briefly, this policy states that all course materials including recordings provided by the course instructor may not be copied, reproduced, distributed or re-posted, unless otherwise given prior written approval by the University. Doing so may be considered a breach of this policy and will be investigated and addressed as possible academic dishonesty, among other potential violations. Improper use of such materials may also constitute a violation of the University's Code of Conduct found at: <https://drexel.edu/studentlife/community-standards/code-of-conduct> and will be investigated as such.

## Recording of Class Activities

In general, students and others should not record course interactions and course activities in lecture, lab, studio or recitation.

Students who have an approved accommodation from the Office of Disability Resources to record online lectures and discussions for note taking purposes should inform their course instructor(s) of their approved accommodation in advance. The recording of lectures and discussions may only be carried out by the students enrolled in the class who have an approved accommodation from Disability Resources with their instructors' prior knowledge and consent. Students with approved accommodations may be asked to turn off their recorder if confidential or personal information is presented.

If a student has any comments, concerns, or questions about provided class materials and/or recording, talk to your course instructor first. If this does not resolve the issue, you can also reach out to the Department Head, and use the process described for a grade appeal to move your concern forward. The process described for grade appeals can be found at: <https://drexel.edu/provost/policies-calendars/student-academic-concerns-complaints>

## Course Schedule and Important Dates

Chapter/section numbers & topics listed in the table below correspond to *Calculus: Early Transcendentals*, 9th/10th/11th edition, by Anton, Bivens, and Davis. As mentioned earlier in the syllabus, you may use this book or you may elect to use a different textbook or electronic resource as a reference.

| Week                     | Material                                                                                                                                                                       | Chapters     | Announcements                                                 |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------------------------------------|
| 1<br>09/21/26            | Area as a Limit & Sigma Notation<br>The Definite Integral                                                                                                                      | 5.4<br>5.5   | Classes begin: Tue<br>09/22                                   |
| 2<br>09/28/26            | The Indefinite Integral<br>The Fundamental Theorem of Calculus                                                                                                                 | 5.2<br>5.6   | Last Day to<br>Add/Drop: Mon<br>09/28                         |
| 3<br>10/05/26            | Integration by Substitution<br>Substitution with Definite Integrals                                                                                                            | 5.3<br>5.9   |                                                               |
| 4<br>10/12/26            | Area Between Two Curves<br>Volumes by Slicing (Disks & Washers)                                                                                                                | 6.1<br>6.2   | Indigenous Peoples'<br>Day (no class):<br>Mon 10/12           |
| 5<br>10/19/26            | Length of a Plane Curve<br>Work                                                                                                                                                | 6.4<br>6.6   | Exam 1: Mon<br>10/19                                          |
| 6<br>10/26/26            | Integration by Parts<br>Integration by Partial Fractions                                                                                                                       | 7.2<br>7.5   |                                                               |
| 7<br>11/02/26            | Improper Integrals<br>Integrating Trigonometric Functions                                                                                                                      | 7.8<br>7.3   | Election Day (no<br>class): Tue 11/03                         |
| 8<br>11/09/26            | Trigonometric Substitutions<br>Separation of Variables                                                                                                                         | 7.4<br>8.2   | Last Day to<br>Withdraw: Mon<br>11/09                         |
| 9<br>11/16/26            | Parametric Equations; Tangent Lines &<br>Arc Length for Parametric Curves<br>Polar Coordinates                                                                                 | 10.1<br>10.2 | Exam 2: Mon<br>11/16                                          |
| Thanksgiving<br>11/23/26 | Catch-up & Review                                                                                                                                                              |              | Thanksgiving<br>Break (no class):<br>Wed 11/25 – Sun<br>11/29 |
| 10<br>11/30/26           | Polar Coordinates<br>Tangent Lines, Arc Length, & Area for<br>Polar Curves                                                                                                     | 10.2<br>10.3 | Classes resume:<br>Mon 11/30                                  |
| Finals Week<br>12/07/26  | The Final Exam Is Cumulative. Date & Time To Be Announced By The Registrar. You are expected to take the final exam at the date and time assigned by the university registrar. |              |                                                               |